Exp NO: 7

Frequency Response of Series and parallel Resonance circuits.

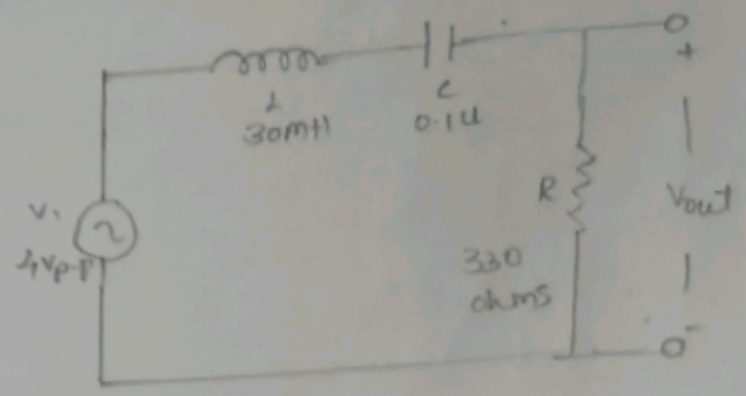
AIM:

To study the behavior of Series and parallel RLC circuits at resonance and to determine the resonant frequency, bandwith and Q of the given RLC circuit using multisim simulator.

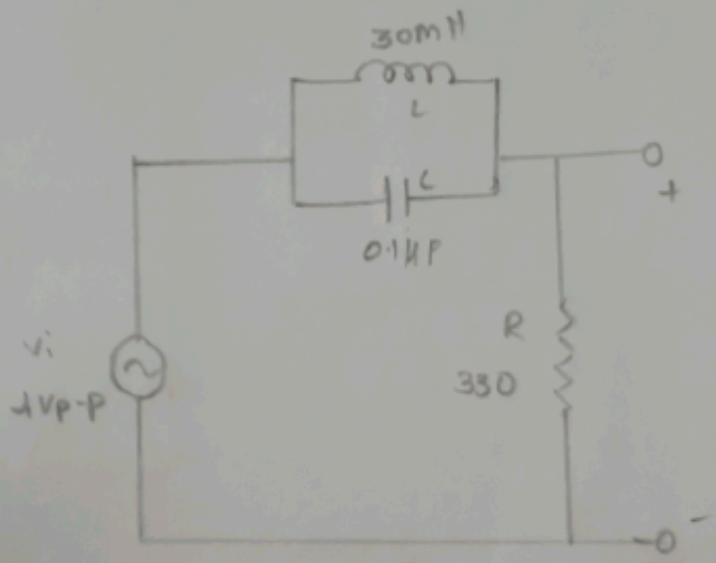
Apparatus Required:

S.NO	Components	Specification	Quantity
1.	CRO	10MHz	1
2.	DIB	-	1
3.	Capacitor	0.1μf	1
4.	Resistor	330Ω	1
5.	Bread board	-	-
6.	Connecting wires	-	-
7.	PC with Multisim Simulator	-	-

Circuit Diagram:



∴ Series RLC circuit



∴ Parallel RLC circuit

formula:

$$f_0 = 1/2\pi\sqrt{LC}$$

$$BW = f_2 - f_1$$

$$Q = f_0 / BW$$

PROCEDURE:

1) Construct circuit shown in fig 1 on bread board

2) Connect CRO ch1 to input and ch2 to output

3) Set the input voltage to 4 Vp-p.

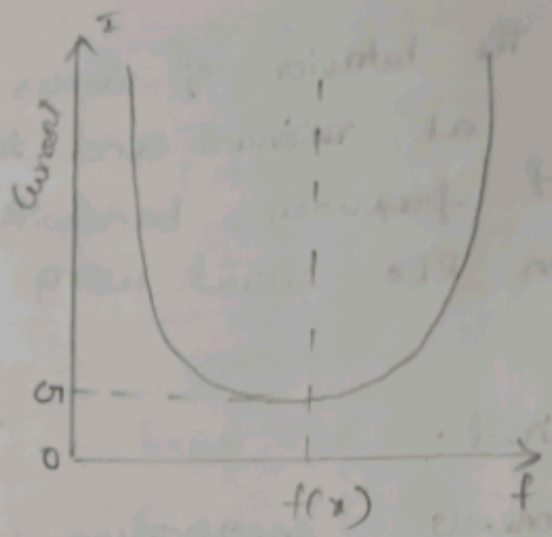
4) Vary the frequency from 500Hz to 3KHz in small steps to get a maximum output voltage magnitude. The frequency at this maximum voltage Vm is the resonance frequency.

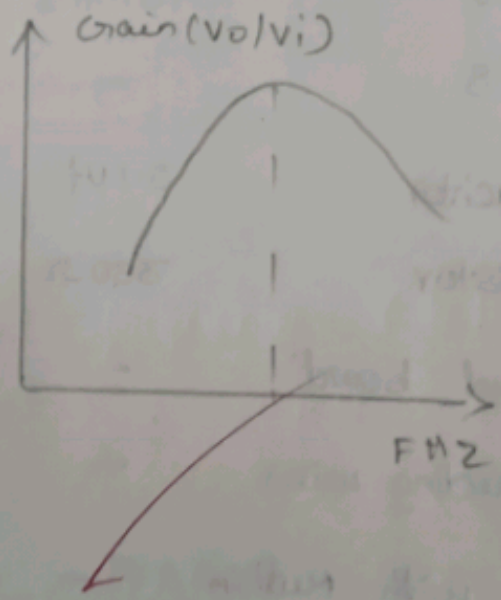
5) If $V_{cutoff} = V_m / \sqrt{2}$, vary the frequency again until the output voltage equals $V_{cut\ off}$.

6) Calculate the bandwith by subtracting the 2 cut-off frequencies.

7) Calculate the Q factor.

Parallel Resonance





Tabulation : $V_{in} = 4V_{p-p}$

Series Resonance			Parallel Resonance		
frequency	output voltage (V)	output current (mA)	frequency (Hz)	output (V)	output current (mA)
500	0.8	2	500	3.8	10
1000	1.6	4	1000	3.2	8
2000	2.8	7	2000	2.4	6
2900	4.0	10	2900	2.0	5
3500	2.5	6	3500	2.7	7
4000	1.5	3	4000	3.4	9

Manual Calculation :

Resonant frequency (Both series and parallel) $L = 30\,mH = 0.03H$, $C = 0.1\,\mu F = 1\times10^{-7}$

$$f_0 = \frac{1}{2\pi\sqrt{LC}} = \frac{1}{2\pi\sqrt{0.03\times10^{-7}}} = 2900\,Hz$$

$$f_0 = 2.9\,KHz$$

Assuming : $f_1 = 2.1\,KHz$, $f_2 = 3.7\,KHz$

Bandwidth : $BW = f_2 - f_1$

$$= 3700 - 2100 = 1600\,Hz$$

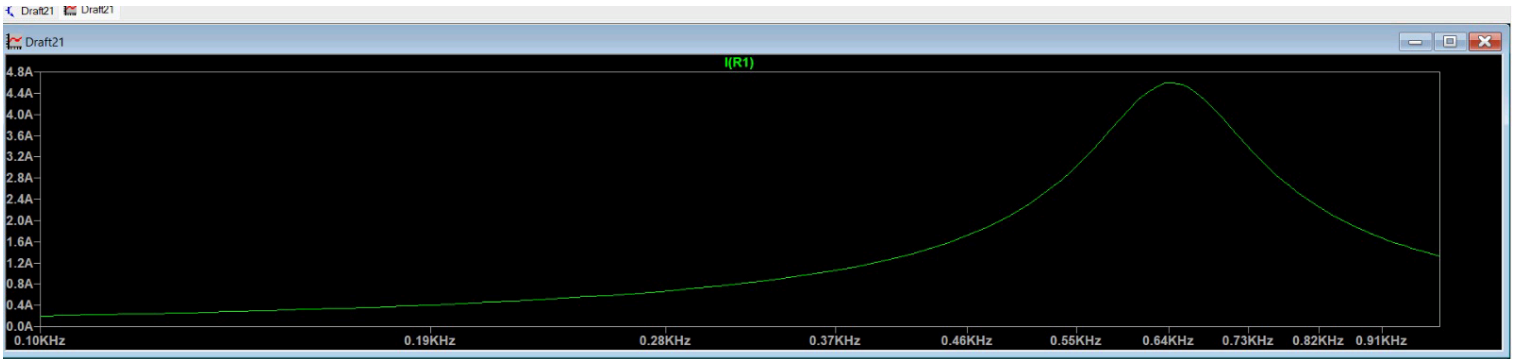
Quality factor (Both series and parallel)

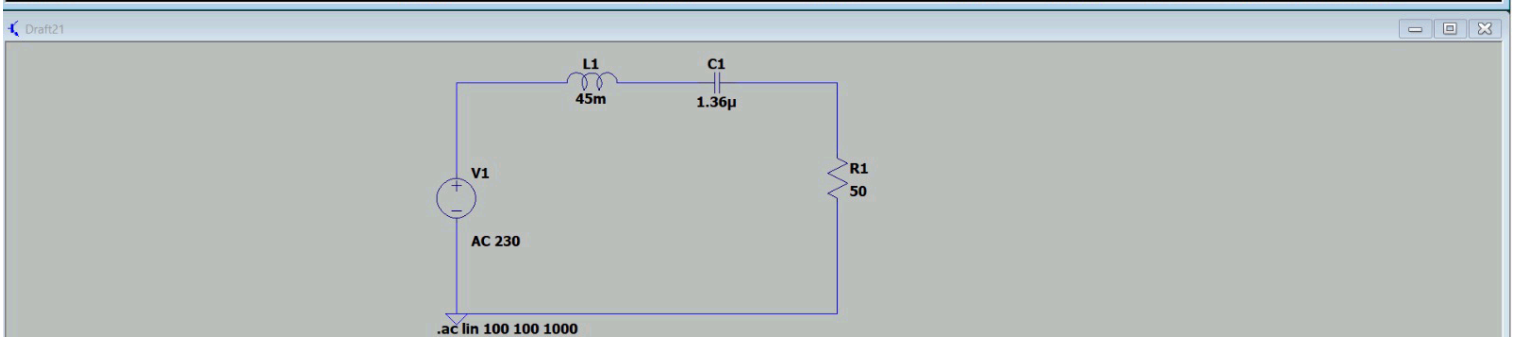
$$Q = \frac{f_0}{BW} = \frac{2900}{1600} \doteq 1.8$$

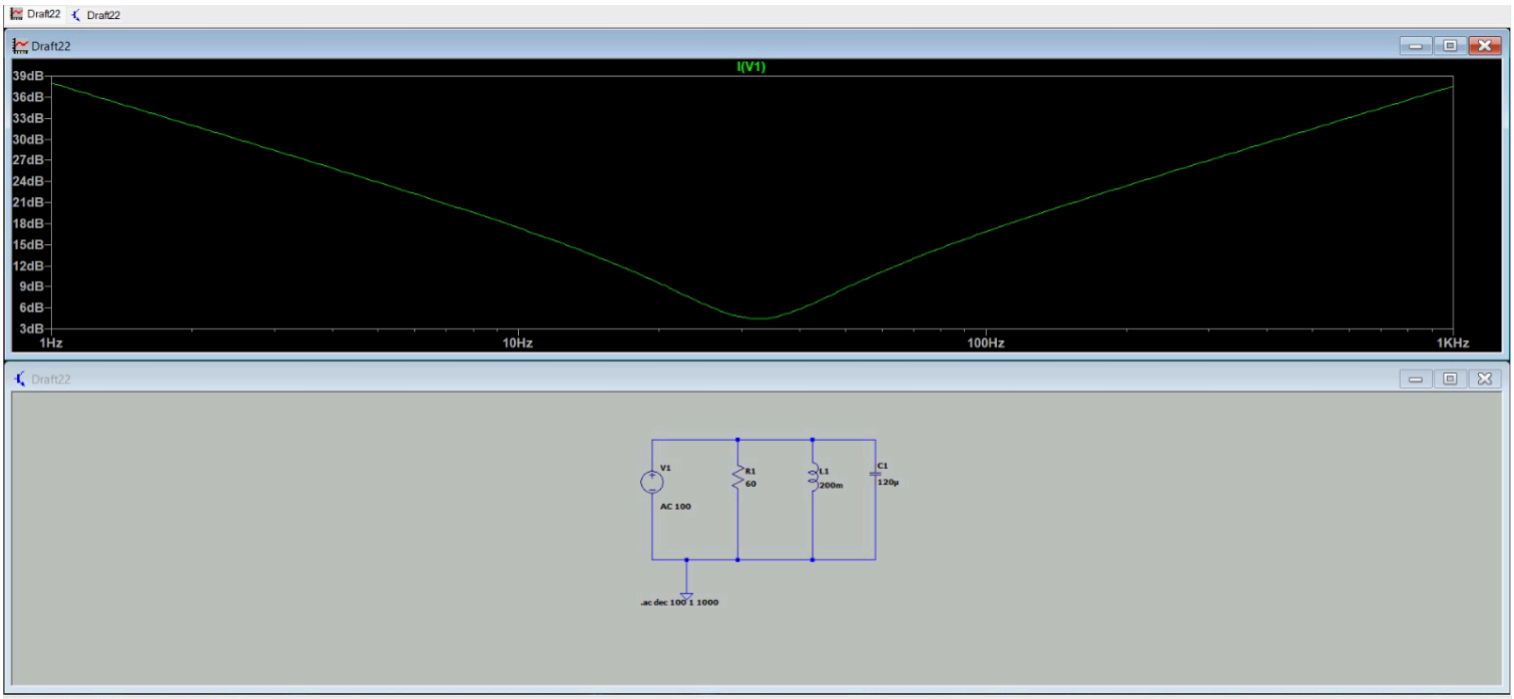
80 / 100 Verified

Marks split-up

Sl.No	Description	Marks Alloted	Marks obtained
1,	Experimental Setup/ connections	20 Marks	20
2,	observation / Readings	20 marks	20
3,	Calculation / Result	20 marks	20
4,	viva	20 marks	20
5,	Record	20 marks	20
		Total:	100


Draft21
I(R1)
4.8A
4.4A
4.0A
3.6A
3.2A
2.8A
2.4A
2.0A
1.6A
1.2A
0.8A
0.4A
0.0A
0.10KHz
0.19KHz
0.28KHz
0.37KHz
0.46KHz
0.55KHz
0.64KHz
0.73KHz
0.82KHz
0.91KHz


Draft21
L1
45m
C1
1.36µ
V1
AC 230
R1
50
.ac lin 100 100 1000



x = 26.718Hz y = 27.708dB

Result:

Series RLC Circuit	Parallel RLC Circuit
Resonant frequency = 2.9 KHz	Resonant frequency = 2.9 KHz
Bandwidth = 1600 Hz	Bandwidth = 1600 Hz
Q factor = 1.81	Q factor = 1.81

100

17/3

Result:

Thus the phenomenon of resonance in RLC circuit was studied and the following were determined using multisim simulator.